

Basic C Logic Concepts for Lab Exam

Prime Number

A prime number has only 2 factors: 1 and itself. Example: 2, 3, 5, 7.

Palindrome

A palindrome remains same when reversed. Example: 121, MALAYALAM.

Armstrong Number

Sum of cube of digits equals original number. Example: $153 = 1^3 + 5^3 + 3^3$.

Perfect Number

Sum of proper factors equals original number. Example: $6 = 1 + 2 + 3$.

Fibonacci Series

Each next number is sum of previous two numbers. Example: 0 1 1 2 3 5 8.

GCD

Greatest Common Divisor is largest number dividing both numbers.

LCM

Least Common Multiple is smallest number divisible by both numbers.

Factorial

Product of numbers from 1 to n. Example: $5! = 120$.

String Reverse

Print characters from end to beginning.

String Length

Count total characters in string before `\0`.

String Compare

Check whether two strings are equal or not.

String Copy

Copy one string into another string.

String Concatenation

Join two strings together.

Sorting

Arrange numbers in ascending or descending order.

Largest in Array

Find biggest element in array.

Special Character Count

Count characters that are not alphabets or digits.

Uppercase and Lowercase Count

Count capital and small letters separately.

Matrix Addition

Add corresponding elements of two matrices.

Matrix Multiplication

Multiply rows of first matrix with columns of second matrix.